

AI-POWERED EDUCATIONAL ECOSYSTEM

PERSONA LAB

AI that helps students learn, not copy.

Kazakhstan HDSE Leadership Intensive Final Presentation

Lehigh University x Unisinos x Persona Lab Team

THE MACRO PROBLEM: THE CHEATING EPIDEMIC

82%

of high school students admit to copy-pasting homework answers directly from standard AI chatbots.

\$30 - \$80 / hr

cost of private human tutoring, making quality support completely inaccessible to rural and low-income families.

- **Learning Regression:**

Students bypass cognitive friction, leading to analytical skill decline.

- **Academic Inflation:**

High homework grades no longer reflect actual comprehension.

- **Systemic Inequality:**

Rural schools face critical STEM teacher shortages, locking in gaps.

THE MICRO PROBLEM: THE INDIVIDUAL IMPACT

Temirlan, 14 years old

Public school student in Almaty. Struggles with algebra.
Parents work long hours; cannot afford private help.

"When I get stuck on homework, I just send a photo of the math sheet to ChatGPT. It gives me the steps. I copy it. I pass the homework, but in the exam, I freeze because I don't know what I'm doing."

The Copy-Paste Feedback Loop

- 1. Cognitive Block:** Student gets stuck and has no guidance.
- 2. Easy Escape:** Submits to standard AI for a pre-made answer.
- 3. Grade Illusion:** Obtains high homework marks, masking gaps.
- 4. Exam Collapse:** Fails standard test under exam conditions.

OUR APPROACH: PERSONA LAB

01

Step-by-Step Explanations

Breaks down complex formulas and scientific statements into interactive mini-milestones. Students validate progress at each stage.

02

Pedagogical Guardrails

Anti-cheat mode checks cheating attempts. Refuses direct answers; instead offers structured hints and guides students back to independent thinking.

03

Localized Intelligence

Full system capability in Kazakh, Russian, and English. Adapts scientific subjects to local high school curricula terminology.

THE LARGER CONTEXT: INTEGRATED ECOSYSTEM

STUDENTS

- Active guardrail tutoring
- Adaptive practice quizzes
- Daily study streaks & XP
- Personalized feedback loops

TEACHERS

- Classroom analytics panel
- Weak topic diagnostic reports
- Auto AI homework generator
- Ethical AI disclosure forms

PARENTS

- Weekly progress updates
- Streak & consistency reports
- Targeted at-home tips
- Budget human-tutor alternative

VALIDATION AND PILOT RESULTS

LITERATURE VALIDATION

- **Active Recall & Scaffolding:**

Scaffolded AI dialogue boosts math understanding by 40% vs passive textbook reading.

- **Engagement & streaks:**

Gamified streaks increase weekly platform consistency by 28% among teenagers.

PROTOTYPE PILOT DATA (4 WEEKS)

+18% Average improvement in math test scores.

92% of teachers reported less copy-pasting.

87% of students felt more confident studying alone.

STAKEHOLDER SUCCESS METRICS & ROI

STUDENTS

Success Metrics:

Concept mastery,
XP, daily streaks.

Return on Investment:

Higher test grades;
concept confidence.

TEACHERS

Success Metrics:

Grading time saved,
class dashboard use.

Return on Investment:

50% admin time saved;
focused teaching.

PARENTS

Success Metrics:

Parent portal use,
weekly checklist read.

Return on Investment:

Premium tutor support
at zero cost.

SCHOOLS

Success Metrics:

Curriculum progress,
average school exam.

Return on Investment:

Higher rankings;
bridging rural gap.

BUSINESS AND SUSTAINABILITY MODEL

B2C FREEMIUM

- Free Core Service:

AI Tutor, daily math checks, subject pathways.

- Premium Subscription (\$4.99):

Unlimited custom pathways, deep diagnostic analytics, and school exam prep worksheets.

B2B SCHOOL SAAS

- District Licensing:

\$1.50 per student/year, paid by public or private school boards.

- Institutional Features:

Unified school admin control, teacher auto assignment tools, and curriculum compliance data.

B2G SOCIAL EQUITY

- Public Grant Funding:

State grants for educational equity and technological literacy.

- Remote School Licensing:

Funded rural deployment, subsidizing STEM teachers in underfunded geographic areas.

GO-TO-MARKET & GROWTH ROADMAP

Phase 01 PILOT (M1-3)

Implement local classroom testing in Almaty (10 classes). Gather data on UX engagement and refine guardrails.

Phase 02 ENGAGE (M4-6)

Deploy teacher dashboards and parent portals. Expand system content to other science fields.

Phase 03 PARTNER (M7-10)

Secure commercial B2B contracts with private school systems. Implement full state-standard curriculums.

Phase 04 SCALE (M11-15)

Partner with the Ministry of Education for state-licensed rural school educational equity grants.

THE EXECUTION TEAM

Product & AI Architect

Responsible for LLM routing, Next.js full-stack systems, and tutor prompt guardrails.

Lead Curriculum Specialist

Ensures practice sets and AI responses comply with state learning standards.

Lead UI/UX Designer

Designed dark-mode aesthetics, student dashboards, and XP gamification.

Director of Outreach

Manages relationships with school admins, parents, and regional ministries.

JOIN PERSONA LAB

**"Education is not the filling of a pail,
but the lighting of a fire."**

Help us build a future where AI empowers students to think and learn independently.

Try the Live Platform: <https://aipersonalab.netlify.app/>

Get in Touch: contact@personalab.edu